From gauge saturation and Weyl modes to exact geometry

23 September 2026. Mathematical continuation of the first-line charter. This note states the new consequences and how their proofs connect. Full arguments are in the four linked manuscripts; earlier sources remain intact. The statements are manuscript theorems, not independent certification of correctness or publication priority.

1. The central realization statement

Consider closed connected oriented smooth four-manifolds with

$$\pi_1(M) \text{ torsion-free}, \qquad \pi_2(M) = 0. \tag{H}$$

This includes aspherical manifolds, but also connected sums of them and additions of $S^1 \times S^3$. The proof of this enlargement is in topological core scope.

Let a be the first Chern class of an actual Spin^c structure solving the unperturbed monopole equations for every metric in $[g]$. Its selection may depend on g; the same structure must work throughout the conformal weights. Set

$$p = (a_g^+)^2, \qquad \delta_-(g) = \| \operatorname{Ric}_g^\sharp + I \|_{\infty,\mathrm{op}},$$

$$q(g,a) = \frac{1}{V_g}\left[\frac{3}{2}\int_M (s_g^2 + 12|W_g^+|^2)\, d\mu_g - 72\pi^2 p\right] \geqslant 0, \tag{1}$$

where Weyl norms are Hilbert–Schmidt curvature-operator norms.

Realization theorem. For fixed $D, A_0 > 0$, there is $\varepsilon(D, A_0) > 0$ such that

$$\boxed{\begin{array}{c} (H), \quad \operatorname{diam}(g) \leqslant D, \quad p \geqslant A_0, \\ \delta_-(g) + q(g,a) < \varepsilon(D, A_0) \\ \Longrightarrow \ \exists (h, J) : \operatorname{Ric}_h = -h, \\ (h, J) \text{ Kähler on the same oriented smooth } M. \end{array}} \tag{2}$$

If $a^2 > 0$, its integral square gives $p \geqslant 1$; then one may use the threshold $\varepsilon(D, 1)$. No prescribed complex structure, cover, independent Weyl bound or negative-square cost is required.

The complete proof uses the preceding rounds in three different roles:

1. The classical scalar monopole inequality supplies noncollapse from $p \geqslant A_0$, the pointwise Ricci bound and bounded diameter.
2. Parameter-two saturation supplies the missing critical concentration control. With $\kappa = \fint |\operatorname{Ric} + g|^2 + q$, every measurable set satisfies

$$\int_E |W^+|^2 \leqslant \frac{4}{3}\operatorname{Vol}(E) + \frac{2}{3}\kappa V. \tag{3}$$

   Thus along a vanishing-defect sequence, no half-flat ALE bubble can survive the topological core obstruction. The earlier harmonic-radius argument gives compactness preserving the smooth type.

3. The inherited actual-form theorem supplies bounded global self-dual forms with vanishing covariant derivative and asymptotic norm $\sqrt{2}$. On the compact limit they become a nonvanishing parallel form, hence an actual global Kähler structure. A fixed small-defect threshold follows by contradiction over changing manifolds.

This closes a previously unproved connection from actual gauge supply to exact realization in a specified regime. It does not claim that a single small average derivative already implies parallelness without compactness. Only first-order form control is used; the old Hessian square-cost budget is not smuggled into (2).

2. A genuine example determines the limit of this implication

The fixed nonzero-shear closed metrics from the second scientific line satisfy, after the normalization $g_\ell = 3G_\ell$,

$$\delta_-(g_\ell) \to 0, \quad V_{g_\ell} = 2\pi^2 A, \quad a^2 = 2\chi + 3\tau = A > 0, \quad p_\ell \geqslant A.$$

Chern–Weil gives the new explicit saturation consequence

$$0 \leqslant q(g_\ell, a) \leqslant 36\delta_-(g_\ell)(1 + \delta_-(g_\ell)) \longrightarrow 0. \tag{4}$$

These are fixed aspherical smooth manifolds with no Kähler structure even on any finite cover. Full curvature and each fixed curvature derivative remain bounded in the inherited construction. They therefore **disprove the diameter-free version of (2)**, not just leave its proof incomplete. The new theorem forces their diameters to diverge.

This is a direct connection between the first line's realization theorem, the second line's actual construction and the third line's virtual-Kähler obstruction. The example does not exclude a completely non-Kähler Einstein metric on that same manifold, so it is not a Yau26 counterexample.

3. Critical chiral information is sufficient for compactness

The critical-threshold theorem provides a second route to realization, without gauge saturation. In (H), fix two-sided Ricci bounds, a diameter upper bound, a volume lower bound and a radius $r_0 > 0$. A uniform small threshold for

$$\sup_x \min\left\{\int_{B_{r_0}(x)} |W^+|^2, \int_{B_{r_0}(x)} |W^-|^2\right\} \tag{5}$$

gives a uniform harmonic radius and topology-preserving compactness. Adding sufficiently small δ_- gives an exact negative Einstein metric on the same smooth manifold. The small half may be different on different balls. This replaces a supercritical L^q, $q > 2$, or vanishing equiintegrability assumption by a fixed critical local threshold.

The corresponding degeneration statement is spatial, not just a global energy inequality: a bubble must carry a definite amount of **both** Weyl energies at the same concentration point. The proof combines a classical Sobolev–Weitzenböck gap on the exact Ricci-flat blowup with core exclusion in both orientations. The explicit threshold and norm conversion are in the proof; no sharp constant is claimed.

Furthermore, the nondegenerate intersection pairing shows that any finite- fundamental-group ALE core embedded in (H) must have zero rational second homology. Thus any remaining smooth ALE bubble would be a rational homology ball with both Weyl halves nonzero. The existence or exclusion of these general mixed-chirality bubbles is not settled by the half-flat classification used here.

4. Exact allowed-mode selection, not another approximate estimate

For an exact $\mathrm{Ric} = -g$ metric let

$$\mathscr{S}(g) = \fint_M (\lambda_3(W^+) - \lambda_2(W^+))^2. \tag{6}$$

The spectral no-bubbling theorem proves that, under (H), bounded diameter and positive volume lower bound, small $\mathscr{S}$ gives full curvature bounds and smooth compactness. There is no pointwise relative pinching or Weyl-amplitude assumption.

Its mechanism is different from simply improving the older constants. On a hypothetical Ricci-flat ALE blowup, vanishing splitting gives the spectrum $(-4t/3, 2t/3, 2t/3)$. The exact Weyl equation gives $\Delta t \leqslant -4t^2$ for the nonnegative Laplacian, and ALE decay forces $t = 0$. The bubble is therefore half-flat and topologically impossible. The nonzero Kähler mode allowed at the negative Einstein scale cannot remain as a nonzero mode at the Ricci-flat ALE scale.

This supplies two exact consequences:

- For every fixed D, an Einstein metric on any $\Sigma_g \times \Sigma_h$, $g, h \geqslant 2$, with diameter at most D and $\mathscr{S} < \varepsilon(D)$, is **locally isometric to** $\mathbb{H}^2_{-1} \times \mathbb{H}^2_{-1}$. Compactness places every counterexample sequence near a zero-splitting limit; the verified complex-product characterization and unobstructed local Einstein moduli then force exact geometry. The finite-cover and eigenline issues are explicitly treated. This is a classification of the metric under test, not merely a new proof that products admit an Einstein metric.
- On the specified nonzero-shear self-gluings, every exact negative Einstein metric with diameter at most D obeys $\mathscr{S} \geqslant \eta(D) > 0$. The previous relative pointwise pinching restriction has been removed. This remains a constraint on possible outputs, not their complete exclusion.

The general allowed-mode statement keeps unobstructedness at KE limits as an explicit condition. It is checked for the product application, not asserted for all negative KE surfaces. The ASD endpoint uses an established local chiral-rigidity theorem. Classical local theorems are credited rather than counted as new results.

5. What this changes in the first-line programme

The original integral obstruction mechanism remains valid independently of the new bounded-diameter regime. The present work adds a precise realization connection and an exact allowed-mode application. Actual gauge saturation, or critical chiral/spectral data, prevents specific bubbles when combined with topology and the global geometric bounds. Smooth-type compactness then becomes available. Global actual forms yield KE realization; an exhausted local Einstein moduli germ yields exact mode rigidity. These outputs have different hypotheses and must not be collapsed into a single unconditional master formula.

The immediate mathematical choices are now sharper. In the compact regime, the remaining concentration must survive the rational-ball and paired-chiral constraints. In the large-diameter regime, the actual shear example proves that saturation cannot alone enforce Kählerization. For its unrestricted Einstein question one needs either a genuinely non-Kähler exact realization or information governing every possible output, not further smallness of the same saturation error. These are specific unresolved alternatives, not a prediction that either will resolve the entire original problem.

No complete sign branch of Yau26 is proved. What is now available is a theorem-level connection among actual fields, topology, degeneration and exact geometry, together with an actual sharpness construction delimiting that connection.